

B. Tech 2nd Semester Sessional Examination-II
Subject-Mathematics-II
Branch- CE (Sec-C), Total Marks-30, Date- 30-04-2025

Answer the following questions:

1. Solve (any three):

5x3=15

a) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = e^x$

b) $\frac{d^2y}{dx^2} + 4y = \cos 2x$

c) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = x$

d) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = e^x \sin x$

e) $\frac{d^2y}{dx^2} + y = \cot x$ (use method of variation of parameters)

f) $\frac{d^2y}{dx^2} + x^2y = 0$ (in power series)

2. Using Cauchy Integral Formula, evaluate **any three** of the integrals:

5x3=15

(a) $\int_C \frac{zdz}{z(z+\pi i)}$ where C is $|z + 3i| = 2$

(b) $\int_C \frac{(5z+1)dz}{z^2+z}$ where C is $|z| = \frac{1}{2}$

(c) $\int_C \frac{z^2 dz}{z^2-4}$ where C is $|z-2| = 1$

(d) $\int_C \frac{10dz}{(z+i)^3}$ where C is $|z+i| = 1$

(e) $\int_C \left(z + \frac{1}{z^2}\right) dz$ where C is $|z| = 2$